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PROJECT AND TEAM INFORMATION

Project Title

Library Management System

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PHASE-II PROJECT PROGRESS DESCRIPTION

Project Abstract

The Library Management System is an academic software project intended to simplify basic library activities. It aims to maintain book information and provide functions for adding and displaying books, searching by title or ID, managing issue and return records, and checking book availability. Data Structures and Algorithms in C can organize and process records, while Object-Oriented Programming concepts in C++ can model library entities and operations. The project demonstrates structured data handling, modular program design, and common library workflows.

Updated Project Approach and Architecture

The system is planned in separate modules:

- **Book Management:** Add, view, update, and search book records.
- **Member Management:** Maintain library-user details.
- **Issue/Return Management:** Record borrowing and returns.
- **Availability Management:** Check and update book status.

Data structures in C can store and traverse records, while C++ classes and objects can represent books, members, and transactions. A menu-driven interface will guide users. Online access and database integration are outside the stated scope unless added later.

Core Functional Requirements

Function	Purpose
Book records	Add, display, update, and search book details.
Member records	Store and retrieve library-member information.
Issue and return	Record borrowing and returns and update availability.
Search and reports	Find books and review availability information.

Tasks Completed

Task / Activity	Responsibility
Identified the problem statement and project scope.	Whole team
Outlined modules for books, members, and issue/return operations.	Shubh Saini, Nilesh Yadav
Selected DSA in C and OOP concepts in C++ as the technology approach.	Whole team
Prepared initial requirements and menu-driven workflow.	Varun Panwar, Nilesh Yadav

Challenges / Roadblocks

A key challenge is selecting suitable data structures so records can be stored, searched, and updated reliably. Another challenge is keeping issue/return operations consistent with book availability. The team should define record formats, separate the program into modules, validate input, and test normal and boundary cases before integration.

Tasks Pending

Task Pending	Responsibility
Implement and connect book add/view/update/search functions.	Shubh Saini
Implement member records and input validation.	Nilesh Yadav
Implement issue/return workflow and availability updates.	Varun Panwar, Shubh Saini
Run integration tests and prepare final demonstration/documentation.	Whole team

Project Outcome / Deliverables

The intended deliverable is a menu-driven Library Management System demonstrating essential library workflows. The planned system will maintain book and member records, support book searches, and record issue/return activities. It also demonstrates data structures, modular programming, and object-oriented design. Expected deliverables include source code, a working demonstration, test cases/results, and project documentation.

Progress Overview

The project scope, core features, and technology approach have been outlined. Next steps are to complete individual modules, connect them through the menu-driven interface, and validate interactions between book records and issue/return operations. A completion percentage is not stated because a verified overall status was not provided.

Expected Benefits

Benefit	Description
Organized records	Keeps book and member information consistent.
Faster lookup	Helps locate book details through search.
Transaction tracking	Records issue/return activity and availability.
Learning outcome	Applies DSA in C and OOP in C++ to a practical problem.

Codebase Information

GitHub Repository:
`https://github.com/nileshyadav17010-arch/Library-management-project-..git`

Working Branch: To be confirmed by the team.

Important Commits: To be documented as development progresses.

Testing and Validation Status

Test Area	Status	Validation Focus
Book management	To be tested	Add, display, update, search, and invalid inputs.
Member management	To be tested	Member entry, retrieval, and validation.
Issue/return operations	To be tested	Availability updates and invalid transactions.
Module integration	Pending	Confirm modules work together through the menu.

Deliverables Progress

- **Project scope and requirements:** Outlined.
- **System architecture:** Core modules identified.
- **Book/member modules:** Implementation and validation to be confirmed.
- **Issue/return module:** Implementation and validation to be confirmed.
- **Integration and testing:** Pending verification.
- **GitHub repository:** Link provided above.
- **Final demonstration and documentation:** To be completed.

Please verify details and project statuses before final submission.